

Harsh Kumar

+91 6376322445 | kum.harshofficial@gmail.com | linkedin.com/in/harsh-kumar-9186b9370 | github.com/thecodeharsh

PROFESSIONAL SUMMARY

Embedded Systems engineer specializing in IoT and PCB design — built a 5-node LoRa mesh network from schematic to field test, winning 1st Prize at ICANCT-2026. Hands-on across ESP32/STM32 firmware, custom PCB layout (KiCad/DipTrace), and hardware debugging, with internship experience at CSIR-CEERI and Indian Railways.

INTERNSHIP EXPERIENCE

CSIR – Central Electronics Engineering Research Institute (CEERI), Jaipur

IoT Intern | Jul 2026 – Aug 2026

- Improved firmware reliability across 5+ embedded prototypes by debugging sensor and wireless communication faults, using Git for version control and serial/JTAG debugging on ESP32, Arduino, and Raspberry Pi platforms.
- Independently designed and built 3 end-to-end ESP32/Arduino touch-display projects — a mini OS, a Wi-Fi network scanner, and interactive games — applying I2C/SPI interfacing and optimizing firmware for resource-constrained hardware.

North Western Railway, Jaipur Division

Summer Intern – Signal & Telecom Department | May 2026 – Jun 2026

- Gained field-level understanding of signal and communication fault diagnosis across the Jaipur division by shadowing engineers on live troubleshooting of RailNet, IPIS, and EI Tower systems over a 45-day placement.

PROJECTS

SNIPS – Scalable Node Interfacing and Processing System

Final Year Project | 1st Prize, ICANCT-2026

- Designed a 5+ node, off-grid LoRa mesh network (Anchor, Relay and Mobile nodes) with a custom PCB and ESP32/Arduino firmware using SX1262 transceivers, addressing the disaster-response communication gap left when cellular networks and GPS fail — operating without gateways or internet unlike standard LoRaWAN.
- Architected a 1-second TDMA superframe (Sync, Control, Data, and Emergency phases) enabling collision-free multi-hop routing and store-and-forward reliability, plus GPS-free positioning via Time Difference of Arrival (TDoA), achieving 15–60m accuracy and over 99% transmission reliability.
- Engineered battery-powered field nodes with Li-ion power regulation/protection circuitry and external SMA/U.FL antennas for range, with Anchor nodes maintaining time sync, the Relay node managing slot allocation, and Mobile nodes running aggressive duty-cycling for field endurance.

Line Follower Robot

- Built a PID-controlled line-following robot on Arduino with a multi-sensor IR array, tuning gains to eliminate oscillation through sharp turns.

Smart Attendance System

- Built a microcontroller-based attendance system that auto-detects and logs student presence via sensor ID, replacing manual roll-call.

TECHNICAL SKILLS

- Languages: C, C++, Python, JavaScript, Dart, HTML, CSS
- Embedded Platforms: ESP32, Arduino, Raspberry Pi, STM32
- Tools: Git, VS Code, Arduino IDE, Proteus, EasyEDA, KiCad, DipTrace, MATLAB, Flutter, AutoCAD (basic)
- Core Concepts: Embedded Systems, IoT, PCB Design, Sensor Interfacing (I2C/SPI/UART), Wireless Comm.
- PCB Design: Designed multiple PCBs in DipTrace — ESP32 dev board, prototyping board, SNIPS comm PCB

EDUCATION

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

Bachelor of Technology in Electronics and Communication Engineering | 2023 – 2027

LEADERSHIP

Ex-Coordinator, OPTICA Student Chapter, SKIT Jaipur

- Coordinated technical workshops and seminars, managing logistics, speaker outreach, and student turnout.

ACHIEVEMENTS

- 1st Prize Winner – ICANCT-2026 (SNIPS – LoRa Mesh Communication System); Top 20 Finalist – HackCrux Hackathon, LNMIIT.